ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

MATHEMATICS

PRACTICE EXAM 1

MARKS: 100 **TIME: 2 hours**

Covers the full Grade 7 year: numbers, exponents, fractions, decimals, percentages, integers, patterns and algebra, geometry, measurement, data handling and probability.

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of NINE questions. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Show ALL your working. Answers alone may not get full marks.
4. Calculators may NOT be used.
5. Diagrams are NOT necessarily drawn to scale.
6. Write neatly and legibly.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 The value of the digit 7 in 4 572 318 is …

A 7 B 700 C 70 000 D 7 000

1.2 Which of these numbers is a prime number?

A 21 B 27 C 29 D 33

1.3 $\sqrt{144} + 2^3 =$ …

A 20 B 14 C 18 D 76

1.4 0,75 written as a common fraction in its simplest form is …

A $\frac{75}{10}$ B $\frac{3}{4}$ C $\frac{7}{5}$ D $\frac{15}{25}$

1.5 $-8 + 5 =$ …

A −13 B 3 C −3 D 13

1.6 The next number in the pattern 3 ; 7 ; 11 ; 15 ; … is …

A 18 B 19 C 20 D 21

1.7 A triangle with all three sides equal is called …

A scalene. B isosceles. C right-angled. D equilateral.

1.8 An angle of 135° is …

A acute. B a right angle. C obtuse. D reflex.

1.9 How many millilitres are there in 2,5 ℓ?

A 25 mℓ B 250 mℓ C 2 500 mℓ D 25 000 mℓ

1.10 The mode of 4 ; 7 ; 2 ; 7 ; 9 ; 4 ; 7 is …

A 4 B 7 C 6 D 5

(10 × 1)

[10]

QUESTION 2: WHOLE NUMBERS

2.1 Round off 38 647 to the nearest: (a) 10 (b) 1 000 (2)

2.2 Calculate. Show ALL your working.

2.2.1 9 004 − 3 678 (2)

2.2.2 347 × 26 (3)

2.2.3 2 856 ÷ 12 (3)

[10]

QUESTION 3: EXPONENTS AND FACTORS

3.1 Write 5 × 5 × 5 in exponential form. (1)

3.2 Calculate: (a) $3^3 + 4^2$ (b) $\sqrt[3]{64} \times \sqrt{25}$ (4)

3.3 Write 18 and 24 as products of prime factors. Use your answers to find the HCF and the LCM of 18 and 24. (4)

[9]

QUESTION 4: FRACTIONS, DECIMALS AND PERCENTAGES

4.1 Calculate. Give your answers in the simplest form.

4.1.1 $\frac{2}{3} + \frac{1}{4}$ (2)

4.1.2 $3\frac{1}{2} - 1\frac{3}{4}$ (3)

4.1.3 $\frac{4}{5} \times \frac{15}{16}$ (2)

4.2 Calculate: (a) 3,45 + 12,7 (b) 0,6 × 0,4 (c) 4,8 ÷ 0,2 (4)

4.3 Arrange from the smallest to the largest: 0,5 ; $\frac{3}{8}$; 45% ; 0,52 (2)

4.4 A pair of school shoes costs R480. During a sale the price is reduced by 25%. Calculate the sale price. (3)

4.5 Thabo scored 42 out of 60 in a test. Write his mark as a percentage. (2)

[18]

QUESTION 5: INTEGERS

5.1 Calculate: (a) −12 + 7 (b) 5 − (−9) (c) −15 − 6 (3)

5.2 In Sutherland the temperature was −4 °C at 05:00 and 13 °C at 14:00. By how many degrees did the temperature rise? (2)

5.3 Arrange from the smallest to the largest: 3 ; −7 ; 0 ; −2 ; 5 (1)

[6]

QUESTION 6: PATTERNS, FUNCTIONS AND ALGEBRA

6.1 Matchsticks are used to make a row of squares, as shown below.



Pattern 1 Pattern 2 Pattern 3

6.1.1 How many matchsticks are needed for Pattern 4? (1)

6.1.2 Write down a rule, in words or as a formula, for the number of matchsticks in Pattern *n*. (2)

6.1.3 How many matchsticks are needed for Pattern 20? (1)

6.1.4 Which pattern uses exactly 100 matchsticks? (2)

6.2 The rule for a flow diagram is $y = 2x + 3$. Copy and complete the table. (3)

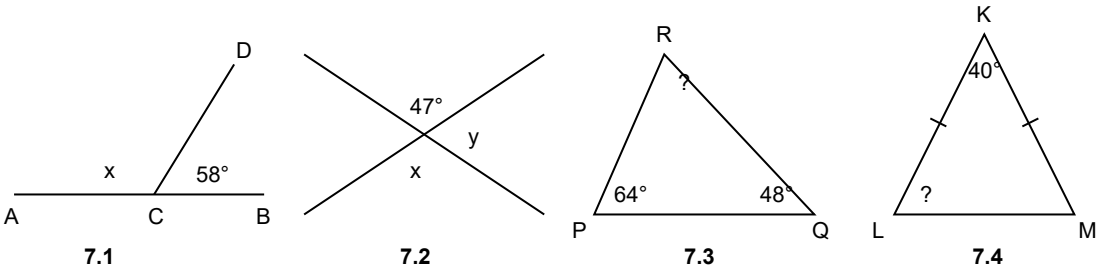
***x* (input)**	1	3	5	(c)
***y* (output)**	5	(a)	(b)	19

6.3 If $a = 4$ and $b = 3$, calculate the value of $5a - 2b$. (2)

6.4 Solve for x by inspection: (a) $x + 8 = 15$ (b) $4x = 36$ (c) $3x - 2 = 13$ (4)

[15]

QUESTION 7: GEOMETRY



7.1 **7.2** **7.3** **7.4**

Diagrams not drawn to scale. In 7.4, KL = KM.

7.1 ACB is a straight line and ∠DCB = 58°. Calculate the size of *x*. Give a reason. (2)

7.2 Two straight lines cross each other. Find the sizes of *x* and *y*. (2)

7.3 In ΔPQR, ∠P = 64° and ∠Q = 48°. Calculate ∠R. Give a reason. (3)

7.4 In ΔKLM, KL = KM and ∠K = 40°. Calculate the size of ∠L. (2)

7.5 Name the quadrilateral that has: (a) four equal sides but no right angles; (b) only one pair of parallel sides. (2)

[11]

QUESTION 8: MEASUREMENT

8.1 A rectangle is 12 cm long and 7 cm wide. Calculate its: (a) perimeter; (b) area. (4)

8.2 Calculate the area of a triangle with a base of 10 cm and a perpendicular height of 6 cm. Use: Area = ½ × base × height. (2)

8.3 A square has a perimeter of 36 cm. Calculate its area. (2)

8.4 A rectangular box (prism) is 5 cm long, 4 cm wide and 3 cm high. Calculate its: (a) volume; (b) total surface area. (5)

[13]

QUESTION 9: DATA HANDLING AND PROBABILITY

9.1 The test marks (out of 20) of 11 learners are: 12 ; 15 ; 9 ; 18 ; 15 ; 11 ; 20 ; 15 ; 14 ; 9 ; 16

9.1.1 Arrange the marks from the smallest to the largest. (1)

9.1.2 Write down the mode. (1)

9.1.3 Write down the median. (1)

9.1.4 Calculate the mean. (2)

9.1.5 Calculate the range. (1)

9.2 A bag contains 3 red, 5 blue and 2 green beads. One bead is taken out without looking. What is the probability that it is blue? Give your answer as a fraction in its simplest form. (2)

[8]

TOTAL: 100

END OF QUESTION PAPER